Practical No. 2

Fragments and Activity Interaction

1. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.
2. Develop an Android application implementing Fragment-based UI navigation between multiple screens.
3. Develop an Android application demonstrating communication between Activity and Fragment.

AIM: Activity Life Cycle, Creating Multiple Activities, Fragment Life Cycle, and Implementing Multiple Fragments in an Android App

THEORY

1. Fragment

A Fragment is a part of user interface within an activity. It can be reused in multiple activities.

Fragment Life Cycle

- onAttach()
- onCreate()
- onCreateView()
- onActivityCreated()
- onStart()
- onResume()
- onPause()
- onStop()
- onDestroyView()
- onDestroy()
- onDetach()

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
```

```
    android:padding="20dp">
    <Button
        android:id="@+id/btnA"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Load Fragment A"/>
    <Button
        android:id="@+id/btnB"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Load Fragment B"
        android:layout_marginTop="10dp"/>
    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Main Activity (MainActivity.java)

```
package com.example.fragmentsdemo;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    Button btnA, btnB;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnA = findViewById(R.id.btnA);
        btnB = findViewById(R.id.btnB);
        btnA.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loadFragment(new FragmentA());
            }
        });
        btnB.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loadFragment(new FragmentB());
            }
        });
    }
    private void loadFragment(Fragment fragment) {
        FragmentManager fm = getSupportFragmentManager();
        FragmentTransaction ft = fm.beginTransaction();
        ft.replace(R.id.fragmentContainer, fragment);
        ft.commit();
```

```
}}
```

2. Fragment A Layout (fragment_a.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is Fragment A"
        android:textSize="24sp"/>
</LinearLayout>
```

Fragment A Java (FragmentA.java**)**

```
package com.example.fragmentsdemo;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class FragmentA extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {
```

```
        return inflater.inflate(R.layout.fragment_a, container, false);
    }}
```

Fragment B Layout (fragment_b.xml**)**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is Fragment B"
        android:textSize="24sp"/>
</LinearLayout>
```

Fragment B Java (FragmentB.java)

```
package com.example.fragmentsdemo;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class FragmentB extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
```

```
                @Nullable Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_b, container, false);
    }
}
```

